How Does the ∼20% Increase in Post-Merger Gravitational-Wave Luminosity from Neutrino Flavor Transformation Affect Future Detection Prospects with Cosmic Explorer and the Einstein Telescope?

A Pedagogical Rewrite of Qiu et al., Phys. Rev. Lett. **135**, 091401 (2025)

For a particle physics audience encountering astrophysics for the first time

Seminar Response

Contents

1 What Is a Neutron Star Merger, and Why Should a Particle Physicist Care? 3
1.1 Neutron Stars in 60 Seconds . . . 3
1.2 The Merger . . . 3
1.3 Why Neutrinos Are Central . . . 3

2 Neutrino Flavor Physics in Extreme Environments 4
2.1 What You Know: Vacuum Oscillations and the MSW Effect . . . 4
2.2 What Is New: The Neutrino Self-Interaction Potential . . . 4
2.3 How Qiu et al. Handle This . . . 5

3 What Is the r-Process and Why Does Y_e Matter? 6

4 The Simulation Setup 6
4.1 The Binary Configuration . . . 6
4.2 The Code: `THC_M1` . . . 6
4.3 Three Models . . . 7

5 Results 8
5.1 Remnant Structure and Composition . . . 8

5.2 Gravitational-Wave Binding Energy and Spectra 9
5.3 Ejecta Composition and Nucleosynthesis Yields 11
5.4 The Subtlety: Where Mixing Occurs Matters 12

6 Gravitational-Wave Detection: A Primer for Non-Astrophysicists 13
6.1 What Detectors Measure 13
6.2 Signal-to-Noise Ratio 13
6.3 Why Post-Merger Detection Is Extremely Hard 14

7 Next-Generation Detectors: Cosmic Explorer and Einstein Telescope 14
7.1 Cosmic Explorer (CE) 14
7.2 Einstein Telescope (ET) 14

8 Answering the Question: Impact of the 20% GW Luminosity Increase on CE and ET 15
8.1 Consequence 1: Enhanced Post-Merger Detectability 15
8.2 Consequence 2: A New Systematic in Equation-of-State Inference 16
8.3 Consequence 3: A Novel Probe of Fundamental Neutrino Physics 16
8.4 Consequence 4: Motivation for Detector Design Choices 17

9 Putting the 20% in Context 17

10 Caveats 17

11 Summary 18

1 What Is a Neutron Star Merger, and Why Should a Particle Physicist Care?

1.1 Neutron Stars in 60 Seconds

A neutron star is the collapsed core left behind after a massive star ($\gtrsim 8\,M_\odot$) exhausts its nuclear fuel and explodes as a core-collapse supernova. The core, no longer supported by radiation pressure from fusion, collapses under gravity until it is stabilized by *neutron degeneracy pressure*—the same Pauli exclusion principle you know from condensed matter, but applied to neutrons at nuclear density. The result is an object with:

- Mass: $\sim$1–2.5 solar masses ($M_\odot \approx 2 \times 10^{30}$ kg)
- Radius: $\sim$10–14 km (roughly the size of a city)
- Central density: $\sim$5–10 times nuclear saturation density ($\rho_0 \approx 2.7 \times 10^{14}$ g/cm^3)
- Surface gravity: $\sim 10^{11}$ times Earth's

Neutron stars are, in a very real sense, giant nuclei—they are the *only* place in the universe where you can study the equation of state (EOS) of cold, dense nuclear matter in bulk. The EOS—the pressure-density relationship $P(\rho)$—encodes the same strong-interaction physics you study with QCD, but in a regime (high density, moderate temperature, nonzero isospin chemical potential) that is inaccessible to lattice QCD and to terrestrial heavy-ion collision experiments.

1.2 The Merger

When two neutron stars orbit each other in a binary system, they lose orbital energy by emitting gravitational waves (GWs)—ripples in spacetime predicted by general relativity. Over millions of years, this energy loss causes the orbit to shrink. In the final minutes, the orbital frequency sweeps upward through the sensitive band of ground-based GW detectors (roughly 10–4000 Hz), producing a characteristic "chirp" signal that increases in both amplitude and frequency. Eventually, the stars collide and merge.

The first direct observation of such an event was **GW170817** on August 17, 2017, detected by the LIGO and Virgo interferometers [1,2]. The source was located in the galaxy NGC 4993 at a distance of $\sim$40 Mpc ($\sim$130 million light-years). It was accompanied by a short gamma-ray burst (detected by the Fermi and INTEGRAL satellites 1.7 seconds after the GW signal) and a "kilonova"—a thermal transient powered by the radioactive decay of freshly synthesized heavy elements in the merger ejecta [2,3].

1.3 Why Neutrinos Are Central

After the two neutron stars collide, the merger remnant is an extremely hot ($T \sim$ 50–100 MeV), dense, rapidly rotating object. At these temperatures, weak-interaction processes

are fast:

$$e^- + p \to \nu_e + n \quad \text{(electron capture)}, \tag{1}$$
$$e^+ + n \to \bar{\nu}_e + p \quad \text{(positron capture)}, \tag{2}$$
$$\nu_e + n \to p + e^- \quad \text{(neutrino absorption)}, \tag{3}$$
$$\bar{\nu}_e + p \to n + e^+ \quad \text{(antineutrino absorption)}. \tag{4}$$

These are all charged-current weak interactions mediated by $W^\pm$ exchange. They set the ratio of protons to neutrons in the material—a quantity called the **electron fraction**:

$$Y_e \equiv \frac{n_p}{n_p + n_n}, \tag{5}$$

where n_p and n_n are the proton and neutron number densities. Pure neutron matter has $Y_e = 0$; equal numbers of protons and neutrons gives $Y_e = 0.5$. The value of Y_e in the material ejected from the merger is the single most important quantity for determining which heavy elements are produced, as we explain in Section 3.

The neutrino luminosity of the remnant is enormous: $L_\nu \sim 10^{53}$ erg/s, comparable to the total luminosity of all stars in the observable universe, sustained for ∼10–100 ms. Neutrinos carry away energy and lepton number, cooling the remnant and shaping its evolution. This is why neutrino physics—including the *flavor* of the neutrinos—matters so much.

2 Neutrino Flavor Physics in Extreme Environments

2.1 What You Know: Vacuum Oscillations and the MSW Effect

From your particle physics courses, you know that neutrinos are produced in flavor eigenstates (ν_e, ν_μ, ν_τ) but propagate as mass eigenstates (ν_1, ν_2, ν_3). The mismatch between these bases, parameterized by the PMNS mixing matrix, causes vacuum oscillations with a characteristic length scale:

$$L_{\rm osc} = \frac{4\pi E_\nu}{\Delta m^2} \sim 10\ \text{km} \times \left(\frac{E_\nu}{10\ \text{MeV}}\right)\left(\frac{10^{-3}\ \text{eV}^2}{\Delta m^2}\right). \tag{6}$$

You also know the MSW (Mikheyev-Smirnov-Wolfenstein) effect: in matter, electron neutrinos experience an additional potential from coherent forward scattering off electrons via W-exchange:

$$V_{\rm MSW} = \sqrt{2}\, G_F\, n_e, \tag{7}$$

where G_F is the Fermi constant and n_e is the electron number density. This potential can cause resonant flavor conversion when $V_{\rm MSW} \sim \Delta m^2/(2E_\nu)$.

2.2 What Is New: The Neutrino Self-Interaction Potential

In a neutron star merger, there is an additional effect that completely dominates the flavor evolution. The neutrino number densities are so enormous ($n_\nu \sim 10^{32}$–10^{33} cm^{-3}) that

coherent forward scattering of neutrinos off *each other*—mediated by Z^0 exchange in the t-channel—produces a potential that dwarfs both the vacuum oscillation term and the MSW potential. This *neutrino self-interaction Hamiltonian* is [4]:

$$H_{\nu\nu}(\mathbf{p}) = \sqrt{2}\, G_F \int \frac{d^3\mathbf{q}}{(2\pi)^3} \, (1 - \hat{\mathbf{p}} \cdot \hat{\mathbf{q}}) \, \big[\varrho(\mathbf{q}) - \bar{\varrho}(\mathbf{q})\big], \tag{8}$$

where $\varrho(\mathbf{q})$ and $\bar{\varrho}(\mathbf{q})$ are the 3×3 density matrices in flavor space for neutrinos and antineutrinos of momentum $\mathbf{q}$. The factor $(1 - \hat{\mathbf{p}} \cdot \hat{\mathbf{q}})$ comes from the current-current structure of the neutral-current interaction.

The critical feature of Eq. (8) is that it is *nonlinear*: the potential experienced by each neutrino depends on the flavor state of every other neutrino. This nonlinearity can drive exponentially growing collective modes called **fast flavor instabilities** [4,5], occurring on timescales of:

$$\tau_{\text{fast}} \sim \frac{1}{\sqrt{2}\, G_F \, n_\nu} \sim \mathcal{O}(\text{ns}). \tag{9}$$

These nanosecond timescales are *seven orders of magnitude shorter* than the hydrodynamic evolution of the merger ($\sim$ms). This enormous scale separation means that (a) flavor conversion happens essentially "instantaneously" from the merger's perspective, and (b) it is computationally impossible to resolve directly in a global simulation.

2.3 How Qiu et al. Handle This

Since resolving nanosecond-scale flavor oscillations within a 30-millisecond merger simulation is infeasible, Qiu et al. [6] model the *outcome* of flavor conversion using a **Bhatnagar-Gross-Krook (BGK) relaxation operator**—a standard technique from kinetic theory that replaces a complicated collision integral with exponential relaxation toward equilibrium:

$$\left.\frac{\partial f_\alpha}{\partial t}\right|_{\text{mixing}} = \frac{1}{\tau_a}\big(f_\alpha^{(a)} - f_\alpha\big), \tag{10}$$

where f_α is the distribution function for neutrino species α, $f_\alpha^{(a)}$ is the target equilibrium distribution, and $\tau_a = 0.4$ ns is the relaxation time [6,7].

The equilibrium state is determined by **detailed balance** for the flavor-exchange reactions:

$$\nu_e\bar{\nu}_e \rightleftharpoons \nu_\mu\bar{\nu}_\mu \rightleftharpoons \nu_\tau\bar{\nu}_\tau, \tag{11}$$

motivated by quantum many-body calculations [8]. Since the simulations track only two effective species—ν_e and ν_x (where x combines μ and τ)—this yields:

$$4\, n_e^{(a)}\, n_{\bar{e}}^{(a)} = n_x^{(a)}\, n_{\bar{x}}^{(a)}, \tag{12}$$

where the factor of 4 accounts for ν_x representing two physical flavors. This is a *lepton-number-preserving* transformation: total lepton number is conserved, but redistributed among flavors.

3 What Is the r-Process and Why Does Y_e Matter?

The **rapid neutron-capture process (r-process)** is responsible for producing approximately half of all elements heavier than iron, including gold, platinum, and uranium [9,10]. The basic mechanism is:

1. Start with seed nuclei (typically iron-group elements, $A \sim 50$–100).
2. In an environment with an enormous density of free neutrons ($n_n \sim 10^{24}$ cm^{-3}), these seeds capture neutrons *much faster than they can β-decay*—hence "rapid."
3. The nuclei become extremely neutron-rich, pushed far from the valley of β-stability toward the neutron drip line (where the neutron separation energy goes to zero).
4. At certain "magic" neutron numbers ($N = 50, 82, 126$), the neutron-capture cross section drops dramatically (closed neutron shells). The r-process path "piles up" at these waiting points.
5. After the neutron supply is exhausted, the unstable, neutron-rich nuclei β-decay back toward stability, producing the observed abundance peaks near $A \approx 80$, 130, and 195.

The crucial requirement is a *very neutron-rich environment*: $Y_e \lesssim 0.25$ [9]. If Y_e is too high (too many protons relative to neutrons), there are not enough free neutrons to drive the rapid capture, and only lighter elements are produced. This is why the neutrino physics matters so directly: neutrino absorption on neutrons (Eq. 3) converts neutrons to protons, *raising* Y_e and *suppressing* the r-process. Anything that reduces the electron neutrino flux—such as flavor conversion—will lower Y_e and enhance heavy-element production.

The observation of the kilonova associated with GW170817 provided the first direct evidence that neutron star mergers are a major site of r-process nucleosynthesis [3,11].

4 The Simulation Setup

4.1 The Binary Configuration

Qiu et al. [6] simulate two identical, non-rotating neutron stars with mass $M = 1.35\,M_\odot$ each, initially separated by 45 km (about 3–4 orbits before merger). The neutron star matter uses the DD2 equation of state [12,13]—a relativistic mean-field model that is "stiff" (predicts relatively large radii $R \sim 13$ km and supports massive remnants). This is important: a stiff EOS ensures the remnant survives for the full ∼30 ms post-merger simulation rather than promptly collapsing to a black hole, giving the neutrino effects time to accumulate.

4.2 The Code: `THC_M1`

The simulations solve three coupled systems simultaneously [6,14]:

1. **Einstein's equations** (in 3+1 decomposition): determines the spacetime geometry.
2. **General-relativistic hydrodynamics**: evolves the fluid (density, velocity, temperature, composition).
3. **M1 neutrino transport**: evolves the zeroth and first angular moments (number density n, energy density E, flux F^α) for four species: ν_e, $\bar{\nu}_e$, ν_x, $\bar{\nu}_x$.

The "M1" stands for "first moment"—it is an approximation to the full Boltzmann transport equation (which lives in 7 dimensions: 3 spatial + 3 momentum + time) that keeps only

the angle-integrated quantities. This makes the problem computationally tractable at the cost of approximating the angular distribution of neutrinos using a closure relation [14,15].

The grid uses 7 levels of adaptive mesh refinement with finest spacing $h \simeq 184$ m, resolving the $\sim$13 km neutron star with $\sim$70 grid points across its diameter.

4.3 Three Models

The three simulations differ only in where the BGK flavor mixing (Eq. 10) is activated [6]:

(i) `No-Mixing`: No flavor conversion anywhere. The baseline.

(ii) **`ρ-11`**: Mixing activated where rest-mass density $\rho < 10^{11}$ g/cm^3. This is the region where *all* neutrino species, including ν_e, are decoupled from matter (optically thin). Physically: flavor conversion only in the free-streaming regime.

(iii) **`ρ-13`**: Mixing activated where $\rho < 10^{13}$ g/cm^3. This extends into the accretion disk interior, where ν_e are still in thermal equilibrium with the medium but heavy-lepton neutrinos (ν_x) can escape freely. Physically: flavor conversion also in the semi-transparent regime.

The density 10^{13} g/cm^3 roughly corresponds to the surface of the remnant [6,16], while 10^{11} g/cm^3 is the decoupling density for electron-type neutrinos.

5 Results

5.1 Remnant Structure and Composition

PHYSICAL REVIEW LETTERS **135,** 091401 (2025)

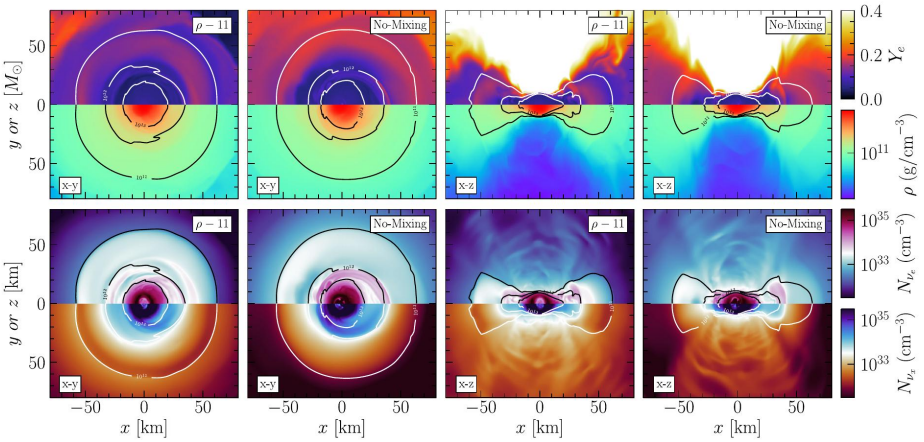


FIG. 1. Simulation snapshots in the equatorial x-y (meridional x-z) planes for the $\rho - 11$ and the `No-Mixing` models at ∼15 ms after merger on the left (right) four panels, respectively. The top-row panels show electron fractions (upper half) and rest mass densities (lower half). The second-row panels display electron neutrino number densities (upper half) and heavy-lepton neutrino number densities (lower half). The contour lines denote where the density is at 10^{11}, 10^{12}, and 10^{13} g/cm^3. We find that after merger, the disk in the $\rho - 11$ mixing model is more neutron rich than that of the `No-Mixing` model. Because of flavor conversion effects transforming electron (anti) neutrinos to heavy-lepton neutrinos, we also find that electron neutrinos are more abundant in the `No-Mixing` than the $\rho - 11$, and vice versa for heavy-lepton neutrinos.

neutrino number densities are chosen such that the reactions $\nu_e\bar{\nu}_e \leftrightarrows \nu_\mu\bar{\nu}_\mu \leftrightarrows \nu_\tau\bar{\nu}_\tau$ are in equilibrium. Since only 4 neutrino species are considered, i.e., $\nu_e, \nu_x, \bar{\nu}_e, \bar{\nu}_x$, where x stands for μ and τ (anti)neutrinos combined, the assumption yields $4n_e^a n_{\bar{e}}^a = n_x^a n_{\bar{x}}^a$, where n is the number densities of neutrinos.

We perform BNS merger simulations with `THC_M1` Neutron star matter is described using the DD2 equation of state [108,109]. We employ 7 levels of adaptive mesh refinement [110,111] using the Carpet driver [112,113] of the Einstein Toolkit [114,115]. The innermost refinement level covers each star prior to merger, as well as the central part of the merger remnant. We present three sets of simulations, (i) no neutrino mixing (`No-Mixing`), (ii) neu-

Figure 1: **From Qiu et al. [6], Fig. 1.** Snapshots at ∼15 ms post-merger. Left four panels: ρ-11 model. Right four panels: `No-Mixing`. Top row: electron fraction Y_e (upper half, red = high Y_e, blue = low Y_e) and rest-mass density (lower half). Bottom row: electron neutrino number density (upper half) and heavy-lepton neutrino density (lower half). Contour lines mark $\rho = 10^{11}$, 10^{12}, 10^{13} g/cm^3. The disk in the ρ-11 model is visibly more neutron-rich (bluer/lower Y_e) than in `No-Mixing`.

Figure 1 is a direct visualization of the neutrino flavor effect on the merger remnant. Focus on the top-row panels: in the `No-Mixing` case (right), the accretion disk surrounding the central remnant shows relatively high Y_e (warmer colors in the upper half). This is because ν_e from the hot remnant are absorbed by neutrons in the disk via $\nu_e + n \to p + e^-$ (Eq. 3), converting neutrons to protons and raising Y_e.

In the ρ-11 case (left), flavor conversion depletes ν_e in the outer disk by converting them to ν_x: the electron neutrino density (bottom row, upper half) is lower, and the heavy-lepton neutrino density is correspondingly higher. With fewer ν_e available for absorption, fewer neutrons are converted to protons, so Y_e stays lower (cooler colors). This "more neutron-rich disk" is the origin of all the downstream nucleosynthesis effects.

5.2 Gravitational-Wave Binding Energy and Spectra

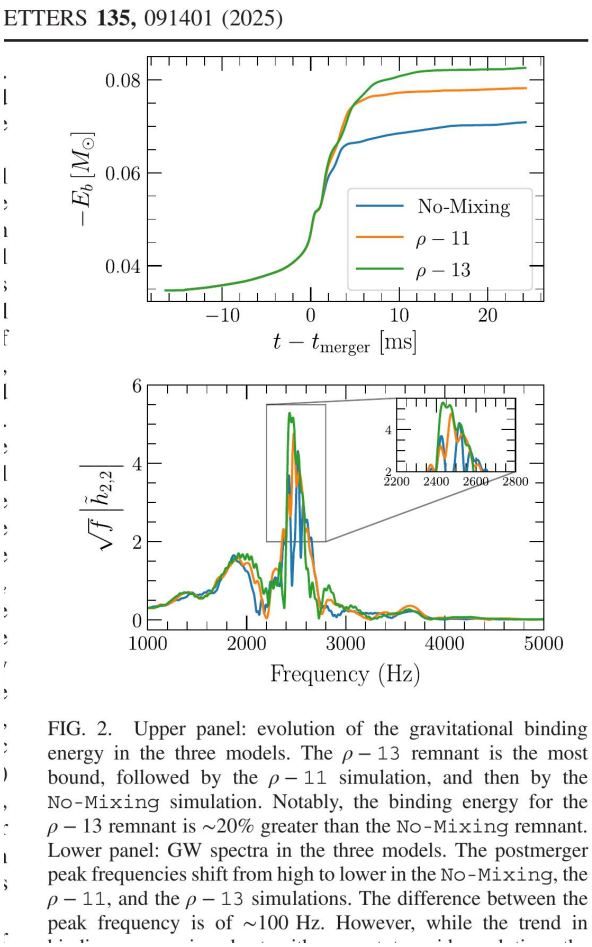


FIG. 2. Upper panel: evolution of the gravitational binding energy in the three models. The $\rho - 13$ remnant is the most bound, followed by the $\rho - 11$ simulation, and then by the No-Mixing simulation. Notably, the binding energy for the $\rho - 13$ remnant is ∼20% greater than the No-Mixing remnant. Lower panel: GW spectra in the three models. The postmerger peak frequencies shift from high to lower in the No-Mixing, the $\rho - 11$, and the $\rho - 13$ simulations. The difference between the peak frequency is of ∼100 Hz. However, while the trend in

Figure 2: **From Qiu et al. [6], Fig. 2.** Upper panel: evolution of the gravitational binding energy $-E_b$ (in solar masses) vs. time relative to merger. ρ-13 is the most tightly bound, then ρ-11, then `No-Mixing`. The ρ-13 binding energy is ∼20% greater than `No-Mixing`. Lower panel: post-merger GW spectra. The inset zooms into the f_2 peak region (∼2200–2800 Hz), showing shifts of ∼100 Hz between models.

This figure contains the key result relevant to the seminar question.

The upper panel of Fig. 2 shows the binding energy of the remnant, computed by integrating the cumulative GW energy losses over time [6]. The physical mechanism creating the ∼20% difference is a chain of cause and effect:

1. Flavor conversion turns $\nu_e/\bar{\nu}_e$ into $\nu_x/\bar{\nu}_x$ in the outer layers of the remnant.
2. ν_x have much smaller interaction cross sections than ν_e. This is because ν_e can interact with matter via both neutral-current (Z^0-exchange) *and* charged-current ($W^\pm$-exchange) processes, while ν_x can only interact via neutral currents. (There are no muons or taus in the medium at these temperatures for $\nu_\mu + n \to \mu^- + p$ to proceed—the muon mass is 106 MeV and the tauon mass is 1.78 GeV, while $T \sim 50$–100 MeV.)
3. Therefore, ν_x escape the remnant much more easily, carrying away energy.

4. The outer layers lose thermal pressure support, contract, and the remnant becomes more compact.
5. A more compact remnant has higher rotational frequency and larger quadrupole moment, radiating GWs more efficiently.

Qiu et al. [6] note that a similar enhanced-cooling effect from flavor conversion was recently reported in core-collapse supernovae by Ehring et al. [17].

The lower panel shows the GW spectrum. The dominant peak, labeled f_2, is the frequency of the fundamental quadrupolar ($\ell = 2$, $m = 2$) oscillation mode of the remnant. It typically falls in the range 2–4 kHz and depends on the EOS and binary parameters [18]. The inset shows that f_2 shifts by $\sim$100 Hz between models. **However, the authors explicitly warn that this frequency shift is not robust:** it was not reproduced at lower resolution and may be a numerical artifact [6]. The binding energy trend, by contrast, *is* robust across resolutions.

5.3 Ejecta Composition and Nucleosynthesis Yields

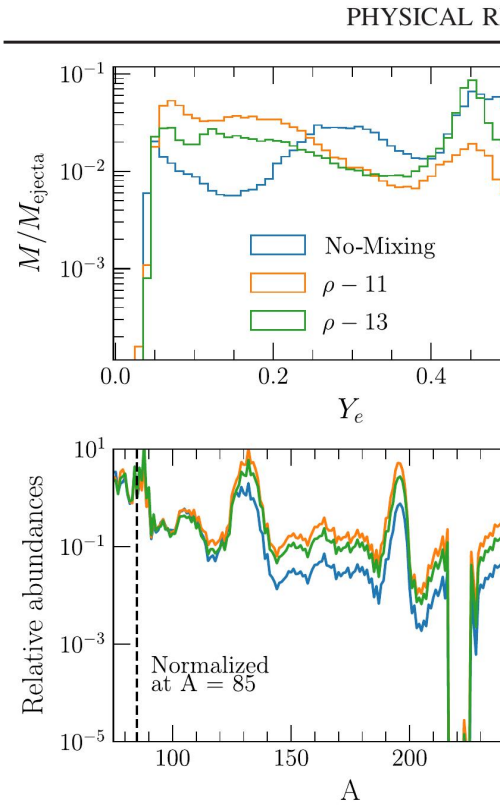


FIG. 3. Upper panel: histograms of the electr distributions of the ejecta. Because of neutrino mixin are generally more neutron rich in the $\rho - 11$ simula $\rho - 13$ simulation than that in the No-Mixing simu $\rho - 11$ simulation has the most low (< 0.2) Y_e ma ejecta, followed by $\rho - 13$ simulation, and then t Mixing simulation. Lower panel: the relative abu nuclei of mass number A formed in the ejecta. We n yields at $A = 85$. Note that the yields for lanthanide elements are one order of magnitude higher in simulation than the No-Mixing simulation. Becaus are determined by the r-process, which will b (suppressed) by more (less) neutron-rich ejecta, between the three models is consistent as the elect histograms.

Figure 3: **From Qiu et al. [6], Fig. 3.** Upper panel: histograms of Y_e in the ejecta. ρ-11 produces the most neutron-rich (low Y_e) ejecta, generating up to 5 times more material with $Y_e < 0.2$ than `No-Mixing` [6]. Lower panel: relative abundances of nuclei as a function of mass number A, normalized at $A = 85$. Lanthanide and heavy-element abundances ($A \gtrsim 120$) differ by up to an order of magnitude between ρ-11 and `No-Mixing` [6].

Figure 3 shows the nucleosynthetic consequences of the neutrino flavor effect. The upper panel is a histogram of Y_e in the unbound ejecta (material escaping the system). The ρ-

11 model has dramatically more low-Y_e (< 0.2) material—the very neutron-rich conditions needed for a vigorous r-process.

The lower panel shows the resulting element abundances after nucleosynthesis (computed at ∼32 years post-merger using precomputed r-process trajectories with the SkyNet code [6,19,20]). The abundances are normalized at $A = 85$ (a light r-process element). For heavier elements—the lanthanides ($A \sim 140$–175, elements like neodymium, europium, gadolinium) and the actinides ($A > 200$, including uranium and thorium)—the ρ-11 simulation produces *an order of magnitude more* than `No-Mixing` [6]. The ρ-13 model falls between, with a factor of ∼2 enhancement. Qiu et al. note that the differences caused by flavor conversion range from approximately 200% to 1000%, far exceeding the 10%–30% numerical uncertainties from transport scheme comparisons [6,21].

These are not small corrections. An order-of-magnitude change in lanthanide abundance would produce a qualitatively different kilonova: lanthanides have enormous bound-bound opacities in the optical/infrared ($\kappa \sim 10$–100 cm^2/g, vs. $\kappa \sim 0.1$–1 cm^2/g for iron-group elements), causing the transient to be redder, dimmer at peak, and longer-lived [3,11].

5.4 The Subtlety: Where Mixing Occurs Matters

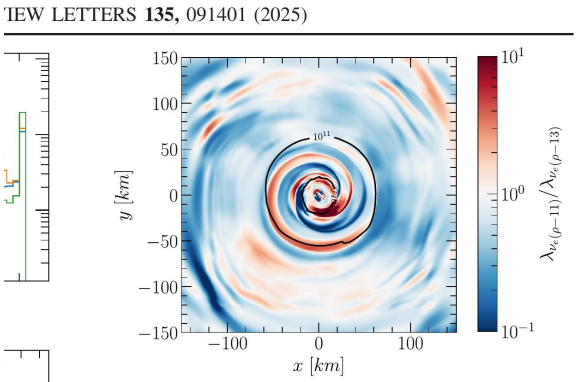


Figure 4: **From Qiu et al. [6], Fig. 4.** Ratio of electron neutrino absorption rates $\lambda_{\nu_e}(\rho\text{-}11)/\lambda_{\nu_e}(\rho\text{-}13)$ in the equatorial plane at ∼15 ms post-merger. Blue regions (ratio < 1) indicate that ρ-13 has *higher* ν_e absorption than ρ-11 in the outer disk. This is caused by "neutrino recycling" (see text).

One of the most interesting results in the paper is that the ρ-11 and ρ-13 models, despite both including flavor conversion, produce quantitatively different ejecta [6]. Figure 4 explains why.

In the ρ-13 model, mixing operates in the disk interior ($10^{11} < \rho < 10^{13}$ g/cm^3). Here, ν_e are still in thermal equilibrium with matter (the medium is optically thick to ν_e), so flavor conversion $\nu_e \to \nu_x$ is immediately compensated by fresh ν_e production via $e^- + p \to \nu_e + n$. The net effect is to create *extra* ν_x that escape the disk (which is optically thin to ν_x). When

these escaping ν_x reach the outer disk ($\rho < 10^{11}$ g/cm^3), some convert back to ν_e, creating a ν_e excess compared to the ρ-11 model [6].

This "neutrino recycling" means that ρ-13 has *higher* ν_e absorption in the outer disk, which raises Y_e and produces less neutron-rich ejecta. The result is that ρ-13 yields up to 3 times less neutron-rich ejecta (and correspondingly fewer heavy elements) than ρ-11 [6].

The rate equation governing this is [6]:

$$\frac{dY_e}{dt} = \lambda_{\nu_e}(1 - Y_e) - \lambda_{\bar{\nu}_e} Y_e \approx \lambda_{\nu_e}, \tag{13}$$

where the approximation holds for neutron-rich material ($Y_e \ll 1$). Higher λ_{ν_e} directly drives faster protonization.

The takeaway: **not only whether flavor conversion occurs, but where it occurs, has a dramatic impact on the nucleosynthesis** [6]. This introduces a new dimension of uncertainty—and opportunity—for multi-messenger astrophysics.

6 Gravitational-Wave Detection: A Primer for Non-Astrophysicists

6.1 What Detectors Measure

A ground-based GW detector is a laser interferometer: a laser beam is split, sent down two perpendicular arms, reflected off mirrors ("test masses"), and recombined. A passing gravitational wave stretches one arm and compresses the other by a fractional amount $h = \Delta L/L$, called the **strain**. For astrophysical sources, $h \sim 10^{-21}$—at Advanced LIGO's 4 km arm length, this corresponds to $\Delta L \sim 4 \times 10^{-18}$ m, roughly 1/1000th the diameter of a proton [1].

For a GW source at luminosity distance D_L radiating power $\dot{E}_{\rm GW}$, the strain scales as:

$$h \propto \frac{1}{D_L}\sqrt{\dot{E}_{\rm GW}}. \tag{14}$$

6.2 Signal-to-Noise Ratio

The detectability of a signal buried in detector noise is quantified by the **matched-filter signal-to-noise ratio (SNR)** [22]:

$$\mathrm{SNR}^2 = 4\int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\,df, \tag{15}$$

where $\tilde{h}(f)$ is the Fourier transform of the strain signal and $S_n(f)$ is the detector's noise power spectral density (PSD). The PSD quantifies how noisy the detector is at each frequency. A signal is conventionally "detected" at SNR $\gtrsim$ 5–8 and "well-characterized" (spectral features measurable) at SNR $\gtrsim$ 8–10.

6.3 Why Post-Merger Detection Is Extremely Hard

The post-merger GW signal has two features that make it difficult to detect:

(a) Short duration: The remnant oscillates for only ∼10–100 ms before either settling down or collapsing to a black hole [18]. Compare this to the inspiral signal, which can last minutes to hours in the detector band. Short signals accumulate less total SNR.

(b) High frequency: The dominant post-merger frequency f_2 falls in the range 2–4 kHz [18], where current detectors are limited by photon shot noise (quantum fluctuations in the number of photons detected). At these frequencies, $S_n(f)$ rises steeply, suppressing the integrand in Eq. (15).

For GW170817 (at $D_L \approx 40$ Mpc), the post-merger signal was *not detected*. The LIGO/Virgo search placed only upper limits, excluding post-merger SNR > 6.7 at 90% confidence [23]. Dedicated matched-filter searches estimate that confident detection would have required the source to be at 1–5 Mpc with current instruments [24]—roughly 10 times closer than it actually was.

7 Next-Generation Detectors: Cosmic Explorer and Einstein Telescope

7.1 Cosmic Explorer (CE)

Cosmic Explorer is a proposed US-based next-generation GW observatory, planned for the mid-2030s to 2040s [25,26]. It was recently recommended for adoption by an NSF subcommittee [26]. The design includes two facilities:

40 km facility: Arms 10 times longer than LIGO's. Since the GW-induced optical signal grows linearly with arm length while most noise sources grow sublinearly, this yields a roughly tenfold improvement in amplitude sensitivity across most of the band [25]. The BNS inspiral detection horizon extends to cosmological distances ($z > 1$) [25].

20 km facility: Specifically designed for post-merger science [25,26,27]. At frequencies above ∼2 kHz, a longer arm can *degrade* sensitivity because the GW half-period approaches the light round-trip time in the cavity:

$$f_{\rm FSR} = \frac{c}{2L} \approx 3.75 \text{ kHz} \quad (\text{for } L = 40 \text{ km}). \tag{16}$$

The 20 km detector can be "tuned" to optimize quantum noise in the 2–4 kHz band, providing ∼30% better post-merger SNR than the 40 km facility in that band [27].

At 100 Mpc with the 40 km CE, post-merger SNRs for optimally oriented sources are ∼5–15, depending on the EOS [28,29]. The BNS merger rate within 100 Mpc is ∼2–3 per year at median rate estimates [28].

7.2 Einstein Telescope (ET)

The Einstein Telescope is a proposed European underground detector with a triangular configuration housing six interferometers (three pairs, each optimized for low or high frequency),

with 10 km arms [30,31]. Its "xylophone" design—pairing a cryogenic low-frequency instrument with a high-power high-frequency instrument—provides broad sensitivity from ∼2 Hz to several kHz [31].

ET is expected to detect $\sim 10^5$ BNS coalescences per year across all redshifts [30]. For nearby events, it will probe the post-merger spectrum. At post-merger SNR $\sim$ 10, the f_2 frequency can be measured to $\mathcal{O}(100)$ Hz precision and the remnant's collapse time to $\mathcal{O}(20)$ ms [32]. ET can also provide early-warning alerts hours before merger thanks to its low-frequency sensitivity [30].

8 Answering the Question: Impact of the 20% GW Luminosity Increase on CE and ET

We now directly address the seminar question. The ∼20% increase in post-merger GW luminosity reported by Qiu et al. [6] has four distinct consequences for Cosmic Explorer and the Einstein Telescope:

8.1 Consequence 1: Enhanced Post-Merger Detectability

From Eq. (14), a 20% increase in GW energy translates to a strain amplitude increase of [6]:

$$\frac{\Delta h}{h} \approx \frac{1}{2}\frac{\Delta E_{\rm GW}}{E_{\rm GW}} \approx \frac{1}{2} \times 0.20 = 10\%. \tag{17}$$

Since SNR $\propto h$ for matched filtering (Eq. 15), this gives a ∼10% boost in post-merger SNR at fixed distance.

The **detection horizon**—the maximum distance at which a given SNR threshold is reached—scales as $D_{\rm horizon} \propto h \propto$ SNR, so:

$$D_{\rm horizon}^{\rm (mixing)} \approx 1.10 \times D_{\rm horizon}^{\rm (no\text{-}mixing)}. \tag{18}$$

The **accessible volume** scales as D^3:

$$\frac{V^{\rm (mixing)}}{V^{\rm (no\text{-}mixing)}} \approx (1.10)^3 \approx 1.33. \tag{19}$$

This means ∼33% more post-merger detections per year [6]. For a 40 km CE with a post-merger horizon of ∼100 Mpc [28,29], the horizon extends to ∼110 Mpc, increasing the event rate from ∼2–3 to ∼3–4 per year at median BNS rate estimates.

For ET, which has comparable high-frequency sensitivity [30,31], the same ∼33% volume increase applies. Given ET's much larger total BNS detection rate ($\sim 10^5$/yr [30]), the subset of nearby events with detectable post-merger signals also grows by ∼33%.

This is a modest but non-negligible improvement. For a signal class that lives right at the detection threshold (SNR $\sim$ 5–10), every additional unit of SNR can make the difference between a non-detection and a measurement.

8.2 Consequence 2: A New Systematic in Equation-of-State Inference

The primary science goal of post-merger detection is measuring the neutron star EOS [18,25,30]. The standard approach uses "quasi-universal relations" that connect the f_2 frequency to the tidal deformability $\tilde{\Lambda}$, which encodes the EOS [18]. At ET sensitivity, f_2 can be measured to $\mathcal{O}(100)$ Hz [32].

Qiu et al. [6] report that neutrino flavor conversion can shift f_2 by up to $\sim$100 Hz—*comparable to the measurement precision of next-generation detectors.* This means:

> **If flavor conversion effects are not included in waveform models, EOS inference from post-merger observations could be systematically biased** [6].

The 20% difference in binding energy (which *is* robust against resolution changes, unlike the frequency shift [6]) also affects the predicted GW amplitude for a given set of binary parameters. An observed post-merger amplitude that exceeds the prediction from inspiral-inferred parameters could be interpreted as evidence for enhanced neutrino cooling via flavor transformation [6,17].

Caveat: The $\sim$100 Hz frequency shift has not yet been confirmed as resolution-independent, and the authors acknowledge it may be a numerical artifact [6]. The binding energy result is on firmer ground.

8.3 Consequence 3: A Novel Probe of Fundamental Neutrino Physics

This is arguably the most exciting implication. The post-merger GW signal becomes sensitive to the outcome of collective neutrino oscillations [6]—a regime of quantum many-body physics involving $\sim 10^{57}$ interacting neutrinos that is:

- Inaccessible to terrestrial neutrino experiments (which measure oscillations of individual neutrinos, not collective effects of $\sim 10^{57}$ of them).
- Inaccessible to lattice calculations (the neutrino self-interaction is nonlinear and out of equilibrium).
- Potentially sensitive to beyond-Standard-Model physics: new neutrino interactions, sterile neutrino mixing, or non-standard self-interactions would change the equilibration rate and outcome [6].

Specifically, Qiu et al. [6] note that the flavor transformation effects "could result from quantum many-body interactions, beyond standard model physics, or flavor instabilities." The ability of CE and ET to detect a $\sim$20% GW luminosity enhancement means they could, in principle, distinguish between these scenarios if the EOS is independently constrained (e.g., from inspiral tidal measurements [1]).

Furthermore, the combination of post-merger GW observations with kilonova photometry from the same event provides a powerful cross-check: the GW signal probes the remnant's compactness (affected by neutrino cooling), while the kilonova probes the ejecta composition (affected by neutrino absorption) [6]. Both are modified by flavor conversion in correlated

ways. This **multi-messenger consistency test** could constrain neutrino flavor transformation independently of the EOS.

8.4 Consequence 4: Motivation for Detector Design Choices

The results strengthen several specific design decisions for CE and ET [6,25,27,30]:

CE's 20 km facility: The importance of the 2–4 kHz band for post-merger science—now amplified by the realization that neutrino physics imprints itself in this band—reinforces the case for this shorter-arm detector optimized for high-frequency sensitivity [25,27].

ET's xylophone configuration: The high-frequency interferometer in each arm pair is designed precisely for post-merger observations [30,31]. The neutrino flavor effects make its science case even more compelling.

Global CE+ET network: A combined network provides broad sky coverage for catching nearby events and the redundancy needed to confidently detect weak post-merger signals, while enabling rapid electromagnetic follow-up for kilonova observations [25,26,30].

9 Putting the 20% in Context

Effect	**Impact on $E_{\rm GW}$ or f_2**	**Robust?**	**Ref.**
EOS variation (all plausible EOSs)	f_2 spans $\sim$1 kHz	Yes	[18]
ν flavor conversion (binding energy)	$\sim$20% in $E_{\rm GW}$	**Yes**	[6]
ν flavor conversion (f_2 shift)	$\sim$100 Hz	**Not yet**	[6]
Hyperon appearance	f_2 shift $\sim$few %	Depends on EOS	[33]
M1 vs. Monte Carlo transport	10–30% in ejecta Y_e	Systematic	[21]
Mass ratio/total mass	Not explored	—	—
Magnetic fields	Not included	—	—

The 20% binding energy effect [6] is larger than estimated numerical uncertainties and comparable to systematic differences between neutrino transport methods [21]. It is a leading-order correction that future simulations and waveform models will need to include.

10 Caveats

- **The BGK model is phenomenological:** It assumes rapid, complete relaxation to a specific equilibrium state [6,7]. Real flavor evolution may not reach this equilibrium, or may reach a different one. Future work using mixing prescriptions calibrated to fast-flavor instability simulations [5,34] is essential.
- **Only one binary configuration:** The $1.35 + 1.35\,M_\odot$ DD2 system is one point in a vast parameter space [6]. Different mass ratios, total masses, and EOSs may enhance or suppress the effect.

- **M1 transport limitations:** The moment scheme approximates the neutrino angular distribution via a closure relation, which introduces systematic errors in semi-transparent regions [6,14,15].
- **Missing physics:** The simulations omit neutrino-lepton inelastic scattering [35], muonic charged-current interactions [36,37] ($\nu_\mu + n \to \mu^- + p$ becomes energetically accessible when $E_\nu > m_\mu \approx 106$ MeV), and magnetic field effects [6]. Each could modify the quantitative results.
- f_2 **shift not converged:** The $\sim$100 Hz frequency shift does not survive resolution checks and may be a numerical artifact [6]. The binding energy result is robust.

11 Summary

Qiu et al. [6] present the first numerical relativity simulations of binary neutron star mergers that include the dynamical effects of neutrino flavor transformation. Their central finding relevant to gravitational-wave astronomy is a $\sim$20% increase in post-merger GW luminosity driven by enhanced neutrino cooling: flavor conversion turns electron neutrinos into heavy-lepton neutrinos that escape more easily, causing the remnant to contract and radiate gravitational waves more efficiently.

For Cosmic Explorer and the Einstein Telescope, this has four consequences:

1. **Modestly enhanced detection prospects:** $\sim$10% more post-merger strain amplitude $\to$ $\sim$33% more accessible volume $\to$ $\sim$33% more post-merger detections per year [6].
2. **A new systematic requiring mitigation:** Flavor conversion effects on the GW spectrum are comparable to the measurement precision of next-generation detectors. Waveform models must include this physics to avoid biased EOS inference [6,32].
3. **A new discovery channel:** Post-merger GW observations, combined with kilonova photometry, could constrain collective neutrino oscillation physics, quantum many-body effects, or BSM neutrino interactions—connecting particle physics to gravitational-wave astronomy in a novel way [6].
4. **Reinforced detector design motivation:** The results strengthen the case for high-frequency-optimized configurations (CE 20 km, ET high-frequency interferometer) and a global CE+ET network [6,25,27,30].

The central message is that neutrino flavor physics is not a small correction but a *leading-order effect* that next-generation gravitational-wave detectors will need to confront—and can potentially exploit as a new window on fundamental physics.

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